

Lakshya JEE 2.0 2025

Maths

DPP: 4

Determinants

Q1 Let $f(x) = \begin{vmatrix} \cos x & \sin x & \cos x \\ \cos 2x & \sin 2x & 2 \cos 2x \\ \cos 3x & \sin 3x & 3 \cos 3x \end{vmatrix}$

Then find the value of $f'(\frac{\pi}{2})$

- (A) 0 (B) 2
(C) 4 (D) 6

Q2 If $f\left(\begin{pmatrix} x \\ x \\ x \end{pmatrix}\right) = \begin{vmatrix} \sin x & \cos x & \sin x \\ \cos x & -\sin x & \cos x \\ x & 1 & 1 \end{vmatrix}$, find the

value of $2f'(0) + \{f'(1)\}^2$

- (A) 0 (B) 1
(C) 2 (D) 3

Q3 Let $ax^6 + bx^5 + cx^4 + dx^3 + ex^2 + fx + g =$
 $\begin{vmatrix} (x+1) & (x^2+2) & (x^2+x) \\ (x^2+x) & x+1 & (x^2+2) \\ (x^2+2) & (x^2+x) & (x+1) \end{vmatrix}$.

Then,

- (A) $f = 3, g = -5$
(B) $f = -3, g = -5$
(C) $f = -3, g = -9$
(D) None of these

Q4 If $f(x), h(x)$ are polynomial of degree 4 and

$$\begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ p & q & r \end{vmatrix} = mx^4 + nx^3 + rx^2 + sx + t$$

be

an identity in x , then

$$\begin{vmatrix} f'''(0) - f''(0) & g'''(0) - g''(0) & h'''(0) - h''(0) \\ a & b & c \\ p & q & r \end{vmatrix}$$

is

- (A) $2(3n - r)$
(B) $2(2n - 3r)$
(C) $3(n - 2r)$
(D) None of these

Q5 If $\begin{vmatrix} x^2 + x & 3x - 1 & -x + 3 \\ 2x + 1 & 2 + x^2 & x^3 - 3 \\ x - 3 & x^2 + 4 & 3x \end{vmatrix} = a_0 + a_1x$
 $+ a_2x^2 + \dots + a_7x^7$.

The value of a_0 is

- (A) 25 (B) 24
(C) 23 (D) 21

Q6 The system of linear equations

$$x + y + z = 2$$

$$2x + y - z = 3$$

$$3x + 2y + kz = 4$$

has a unique solution, is

- (A) $k \neq 0$
(B) $-1 < k < 1$
(C) $-2 < k < 2$
(D) $k = 0$



Q7 If the trivial solution is the only solution of the system of equations

$$\begin{aligned}x - ky + z &= 0 \\ kx + 3y - kz &= 0 \\ 3x + y - z &= 0\end{aligned}$$

Then, the set of all values of ' k ' is:

- (A) $\{2, -3\}$
(B) $\mathbb{R} - \{2, -3\}$
(C) $\mathbb{R} - \{2\}$
(D) $\mathbb{R} - \{-3\}$

Q8 If the system of equations $x - 2y + 5z = 3$, $2x - y + z = 1$ and $11x - 7y + pz = q$, has infinitely many solutions, then

- (A) $p + q = 2$
(B) $p + q = 10$
(C) $p - q = 2$
(D) $p - q = 5$

Q9 If the system of equations

$$\begin{aligned}x + 4ay + az &= 0 \\ x + 3by + bz &= 0 \\ x + 2cy + cz &= 0\end{aligned}$$

have a non-zero solution, then $a, b, c (\neq 0)$ are in

- (A) AP (B) GP
(C) HP (D) AGP

Q10 The system of equations

$$\begin{aligned}ax - y - z &= a - 1, x - ay \\ -z &= a - 1, x - y - az &= a - 1\end{aligned}$$

has no solution, if a is

- (A) either -2 or 1
(B) -2
(C) 1
(D) not -2

Q11 If $0 \leq \theta \leq \pi$ and the system of equations

$$\begin{aligned}x &= (\sin\theta)y + (\cos\theta)z \\ y &= z + (\cos\theta)x \\ z &= (\sin\theta)x + y\end{aligned}$$

has a non-trivial solution, then $\frac{8\theta}{\pi}$ is equal to

- (A) 2 (B) 4
(C) 6 (D) 8

Q12 Given $2x - y + 2z = 2$, $x - 2y - z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equation has no solution is:

- (A) -3 (B) 1
(C) 0 (D) 3

Q13 The system of equations $ax + 4y + z = 0$, $bx + 3y + z = 0$, $cx + 2y + z = 0$ has non-trivial solution if a, b, c are in

- (A) AP
(B) GP
(C) HP
(D) None of these

Q14 If t is a real number and $k = \frac{t^2 - t + 1}{t^2 + t + 1}$, then the system of equations $3x - y + 4z = 3$; $x + 2y - 3z = -2$ and $6x + 5y + kz = -3$ for any allowable value of k , has

- (A) a unique solution
(B) infinite solutions
(C) no solution
(D) 2 solutions

Q15 The number of values of k for which the system of equations

$$(k + 1)x + 8y = 4k, kx + (k + 3)y = 3k - 1$$

- has infinitely many solutions, is
(A) 0 (B) 1
(C) 2 (D) infinite

Q16 Let (x, y, z) be points with integer coordinates satisfying the system of homogeneous equations:



$$3x - y - z = 0$$

$$-3x + z = 0$$

$$-3x + 2y + z = 0$$

Then the number of such points for which

$$x^2 + y^2 + z^2 \leq 100 \text{ is}$$

(A) 1 (B) 3

(C) 5 (D) 7

Q17 The number of all possible values of θ , where $0 < \theta, < \pi$, for which the system of equations

$$(y + z)$$

$$\cos 3\theta = (xyz) \sin 3\theta$$

$$, x \sin 3\theta = \frac{2 \cos 3\theta}{y} + \frac{2 \sin 3\theta}{z}$$

$$, (xyz) \sin 3\theta = (y + 2z) \cos 3\theta + y \sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is

(A) 1 (B) 2

(C) 3 (D) 4



Answer Key

Q1 (C)
Q2 (D)
Q3 (D)
Q4 (A)
Q5 (D)
Q6 (A)
Q7 (B)
Q8 (C)
Q9 (C)

Q10 (B)
Q11 (C)
Q12 (D)
Q13 (A)
Q14 (A)
Q15 (B)
Q16 (D)
Q17 (C)



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